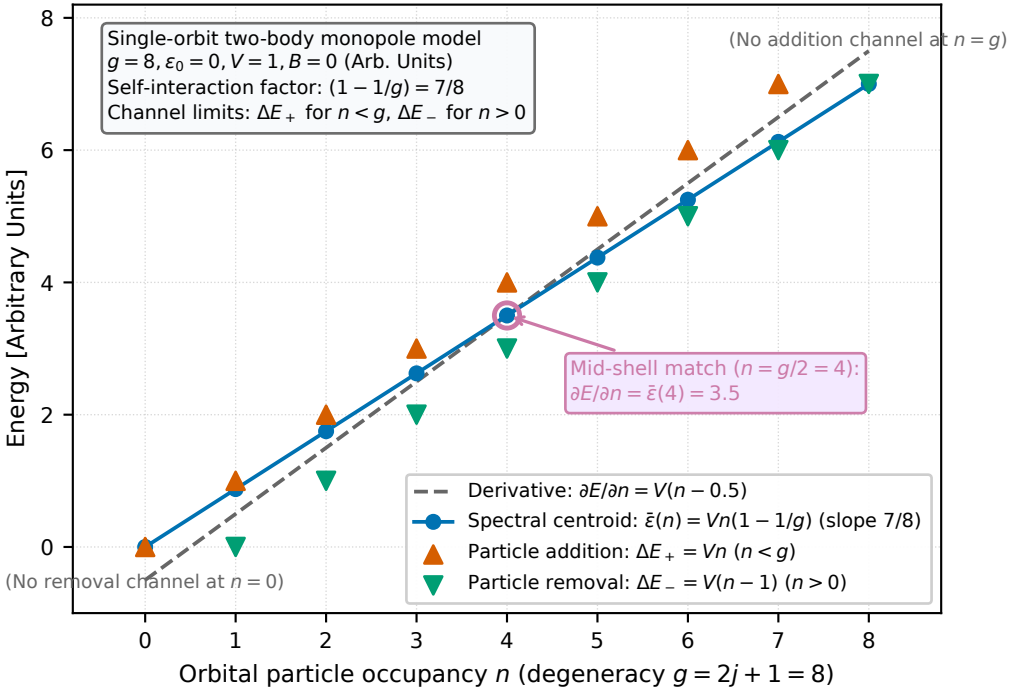
Single-orbit two-body monopole model
$g = 8, \varepsilon_0 = 0, V = 1, B = 0$ (Arb. Units)
Self-interaction factor: $(1 - 1/g) = 7/8$
Channel limits: ΔE_+ for $n < g$, ΔE_- for $n > 0$
(No addition channel at $n = g$)
Mid-shell match ($n = g/2 = 4$):
$\partial E/\partial n = \tilde{\varepsilon}(4) = 3.5$
Derivative: $\partial E/\partial n = V(n - 0.5)$
Spectral centroid: $\tilde{\varepsilon}(n) = Vn(1 - 1/g)$ (slope 7/8)
Particle addition: $\Delta E_+ = Vn$ $(n < g)$
Particle removal: $\Delta E_- = V(n - 1)$ $(n > 0)$
(No removal channel at $n = 0$)
Energy [Arbitrary Units]
Orbital particle occupancy n (degeneracy $g = 2j + 1 = 8$)
0
1
2
3
4
5
6
7
8